

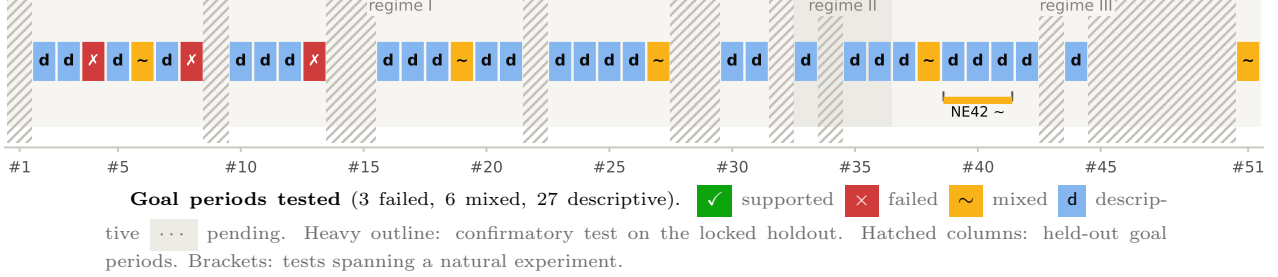
H91: Eigenvector rotation as a reorganization signal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

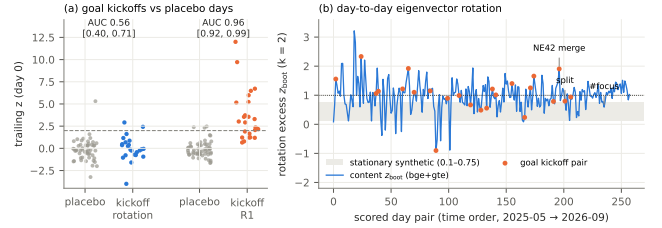
Bottom line. Collective-mode rotation does not mark goal changes; the content modes drift slowly every day, and only a room merge moves them beyond noise.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test asks whether the collective modes of the agent content matrix turn at reorganizations. For each pair of consecutive active days it measures how far the top two eigenvectors rotate, relative to the finite-sample noise of each day. It runs on 258 non-holdout day pairs in content (both embedding models) and 237 in talk. The rotation alarm does not flag goal kickoffs: AUC 0.56 [0.40, 0.71] on 23 kickoffs, against 0.96 for H36's centroid shift R1. It catches none of R1's 13 misses. Between events the content modes stay within the noise on 86% of quiet pairs. But every day rotates about one standard deviation more than a stationary swarm, with no spike at events. The NE42 room merge is the one event where talk and content both rotate ($p \leq 0.01$). The holdout script exists and is not run.



Observables. (a) Alarm scores on day 0 of 23–24 goal kickoffs (color) and on 51 placebo days (gray). The rotation alarm overlaps the placebos; R1 separates them. (b) Rotation excess z for every scored day pair in time order. It sits near 1 throughout, above the band of stationary synthetic swarms (gray). Kickoffs (orange) do not stand out. The NE42 merge is the largest content excess among room events.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Inverse Ising and vector spins (models 01, 11) with random-matrix theory. A content spin is an agent's 30-min window vector, centered on its own day. The day overlap matrix Q_d has eigenvectors v_k : its collective modes. v_1 is the uniform mode. The rotation δ_2 is the subspace distance between two days' top-2 modes. The null is finite- T eigenvector diffusion (the Dyson picture), estimated by block-bootstrapping each day. A is a trailing alarm in units of the recent spread.

$$Q_{ij}(d) = \frac{1}{W_d D} \sum_w \mathbf{z}_i(w) \cdot \mathbf{z}_j(w), \quad Q v_k = \lambda_k v_k$$

$$\delta_2(a, b) = \left[1 - \frac{1}{2} \|V_a^\top V_b\|_F^2 \right]^{1/2}, \quad z = \frac{\delta_2 - \langle \delta_2 \rangle_{\text{boot}}}{\sigma_{\text{boot}}}$$

$$\langle \sin^2 \theta_1 \rangle_{\text{Dyson}} \approx \frac{2}{T} \sum_{j>1} \frac{\lambda_1 \lambda_j}{(\lambda_1 - \lambda_j)^2}, \quad A(d) = \frac{z(d) - \text{med}_{10} z}{\text{sd}_{10} z}$$

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 synthetic size and power	size ≤ 0.05 ; power 0.2–0.4 at N 8–16, W 8; ≥ 0.85 at W 16	partial
P1 kickoff AUC ≥ 0.70 , not worse than R1	0.56 [0.40, 0.71]; R1 0.96; Δ -0.41 [-0.56 , -0.27]	fail
P2 Dyson between events, $s_{\text{sig}} \leq 0.15$	content 0.14 (bge 0.20, gte 0.08); talk 0.17 (n 6)	content pass
P3 talk $z \geq 2$ at NE42	merge 1.97 (p 0.005); split 1.47 (p 0.025)	fail (thresh-old)
P4 complement, $\rho(A, R1) < 0.3$	0.21	pass
P5 model robustness	bge–gte ρ 0.33; style variant AUC +0.09	fail
P6 not composition	$\rho(z, \Delta N)$ 0.01–0.05	pass
N1 NE42 A-B-A	merge rotates talk and content; split weaker	mixed
N2 #51 baseline, #focus	content within noise (0.13); talk 0.40; #focus silent	mixed

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\times$ failed
G05 d descriptive
G06 $\sim$ mixed
G07 d descriptive
G08 $\times$ failed
G10 d descriptive
G11 d descriptive
G12 d descriptive
+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

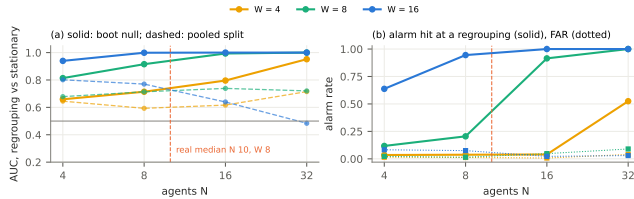
Most kickoff pairs are small ($N \approx 10$, 8 windows), where a strong regrouping is caught only 20–40% of the time. No kickoff fell in the powered range, so P1 cannot rule out rotation at strong regroupings. The null was changed once, after the synthetic study and before real data. Per-day rotation agrees only weakly between embedding models. The talk null is anti-conservative at $N \leq 8$. 27 of 35 periods have too few quiet pairs for a per-period verdict.

ROUND 2

Test rotation at operator room assignments and in 8-h, $N \geq 16$ periods (held-out #45–#50). Fit the daily drift as a rotational diffusion constant per period and place it on the phase diagram.

- **Essence.** whether the swarm’s co-movement structure has events at all, or only a slow daily drift that a monitor must treat as its baseline.
- **H91-R1** Test rotation where it is powered: operator room assignments (DQ6) and 8-h, $N \geq 16$ periods (#51 era, held-out #45–#50), not goal kickoffs.
- **H91-R2** Time the NE42 merge rotation with within-day halves, to see whether the modes turn at the merge hour.
- **H91-R3** Fit the daily drift ($z \approx 1$) as a rotational diffusion constant per period and use it as a phase-diagram coordinate.

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Second figure. Synthetic validation at village counts: a regrouping of half the agents (two rooms, loading 0.4, window persistence 0.5). (a) AUC against stationary days for the same-day bootstrap null (solid) and the pooled-split null (dashed). The split null loses power because it mixes the two days. (b) Trailing-alarm hit rate (solid) and false-alarm rate (dotted). The orange line marks the median real day.